BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Strings and Input Checking

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What is the result of concatenating 'AB' and '12'?

2. How is an item accessed in a 1D array?
   A) Using its font
   B) Using a file extension
   C) Using a network address
   D) Using an index

3. "Testing a few inputs proves a program can never contain any fault." Correct this. Explain why.

4. A programmer tests input code while building a booking program, then tests the finished program. Correct both claims. Explain why. (a) Iterative testing starts only after delivery. (b) Final testing removes the need for earlier tests.

Today's objectives

Manipulate strings and check inputs against rules. Use the technique in an independent scenario.

Key vocabulary - Read, cover, recall

Length check

A string's .length property can check that a minimum number of characters were supplied.

Validation

Checking entered data against rules improves reliability but does not verify identity.

Input checking

Reject a value that fails a stated rule and request another valid one.

Authentication

Checking a user's identity is different from checking whether input fits a format.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Length check
A string's .length property can check that a minimum number of characters were supplied. Apply this in a school portal username.

Validation
Checking entered data against rules improves reliability but does not verify identity. Apply this in a library account.

Input checking
Reject a value that fails a stated rule and request another valid one. Apply this in a club membership.

Authentication
Checking a user's identity is different from checking whether input fits a format. Apply this in a mobile app sign-up.

I Do - Worked model

A school portal username is being developed. Explain how length check and validation are used. [4 marks]

- Read the scenario and identify the data the program needs.
- Length check [1]; A string's .length property can check that a minimum number of characters were supplied. [1].
- Validation [1]; Checking entered data against rules improves reliability but does not verify identity. [1].
- Two linked points for each concept.

We Do - Guided practice

1. A library account uses length check. Explain how it helps. [2 marks]

2. Explain validation in a club membership. [2 marks]

3. A mobile app sign-up uses input checking and authentication. Explain both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Replace the fixed password with an input instruction.

```
password = "abc123"
if password.length >= 6 then
print("Length OK")
else
print("Too short")
endif
```

You Do - Independent exam questions

Q2. State TWO relevant techniques from this topic. [2 marks]

Q3. Explain how validation and input checking help with a video service username. [4 marks]

Q4. A programmer designs a sports team password. Explain why length check, input checking and authentication matter. [6 marks]

Exit Ticket - True or False?

A string's .length property can check that a minimum number of characters were supplied.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. AB12, with no automatically inserted space.

2. D - Using an index

3. To identify faults and check behaviour against requirements using expected results.

4. (a) Iterative testing checks and fixes parts while building the program. (b) Final testing checks the finished program against its requirements. Both are needed.

I Do - Explain the model

Read the scenario and identify the data the program needs.

Length check [1]; A string's .length property can check that a minimum number of characters were supplied. [1].

Validation [1]; Checking entered data against rules improves reliability but does not verify identity. [1].

Two linked points for each concept.

We Do - Model answers

1. Length check [1]; A string's .length property can check that a minimum number of characters were supplied. [1].

2. Validation [1]; Checking entered data against rules improves reliability but does not verify identity. [1].

3. Input checking [1]; Reject a value that fails a stated rule and request another valid one. [1]. Authentication [1]; Checking a user's identity is different from checking whether input fits a format. [1].

Code First - Answer

```
password = input("Test password:")
if password.length >= 6 then
print("Length OK")
else
print("Too short")
endif
```

Expected output: Length OK (if student enters abc123)

You Do - Self-mark

Q2: Length check [1]; Input checking [1].

Q3: Validation [1]; Checking entered data against rules improves reliability but does not verify identity. [1]. Input checking [1]; Reject a value that fails a stated rule and request another valid one. [1].

Q4: Length check [1]; A string's .length property can check that a minimum number of characters were supplied. [1]. Input checking [1]; Reject a value that fails a stated rule and request another valid one. [1]. Authentication [1]; Checking a user's identity is different from checking whether input fits a format. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.